2024 AMC 10A Problem 15 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10A Problem 15. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10A Problem 15

Problem:
Let M be the greatest integer such that both $M+1213$ and $M+3773$ are perfect squares. What is the units digit of M ?

Answer Choices:

A. 1

B. 2

C. 3

D. 6

E. 8

Solution:

Suppose $M+1213=j^2$ and $M+3773=k^2$ for nonnegative integers j and k. Then

$$(k+j)(k-j)=k^2-j^2=3773-1213=2560=5\cdot 2^9$$

Because $k+j$ and $k-j$ have the same parity and their product is even, they must both be even, and it follows that one of them is $5\cdot 2^i$ and the other is 2^{9-i} for some i with $1\leq i\leq 8$. Solving for k gives

$$k = \frac{5 \cdot 2^i + 2^{9-i}}{2}$$

To maximize M it is sufficient to maximize k, and this will occur when $i = 8$ and $k = 5 \cdot 2^7 + 1 = 641$. Therefore $M = 641^2 - 3773$, and its units digit is $(\mathrm{E})\boxed{8}$.

OR

Because $(n+1)^2 - n^2 = 2n + 1$, successive terms in the sequence of squares, $1, 4, 9, 16, \ldots$, differ by successive odd numbers; and because $(n+2)^2 - n^2 = 4(n+1)$, the terms in this sequence that are two apart differ by successive multiples of 4 . The two squares required in this problem differ by $3773 - 1213 = 2560$, a multiple of 4 . It follows that the greatest such squares are two apart in the sequence of squares, so $n + 1 = \frac{2560}{4} = 640$. Therefore these squares are $n^2 = 639^2$ and $(n+2)^2 = 641^2$, and $M + 1213 = 639^2$. Then $M = 639^2 - 1213$, and its units digit is $(\mathrm{E})\boxed{8}$.

Note: Shown below is a table of $5 \cdot 2^i$, 2^{9-i}, k, j, and M for each i (notation from the first solution). Observe that M is maximized when k is maximized.

i	$5 \cdot 2^i$	2^{9-i}	k	j	M
1	10	256	133	123	13916
2	20	128	74	54	1703
3	40	64	52	12	-1069
4	80	32	56	24	-637
5	160	16	88	72	3971
6	320	8	164	156	23123
7	640	4	322	318	99911
8	1280	2	641	639	407108